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Electrical power connector assembly

Abstract

A spacer for maintaining one or more wires during a molding process, the spacer having an external frame, and a plurality of spring curved leaves. Each spring curved leaf has a fixed end attached to an inner face of the external frame and a free end. The free ends of the plurality of spring curved leaves are arranged to flexibly wrap the one or more wires.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to European Patent Application No. 22172204.4 filed on May 6, 2022, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD OF THE INVENTION

(2) The disclosure generally relates to connector assemblies for automotive vehicles. More particularly, this disclosure relates to a connector assembly for electrically connecting electrical conductors carrying high power currents.

BACKGROUND

(3) An example of such a connector assembly is disclosed by US Patent Application Publication 2018/0375227 A1. When connectors are used in high voltage applications, e.g., greater than 48 volts, it is desirable to prevent accidental contact with energized terminals or conductors. More particularly, access to an electrical conductor through a cavity or passage of an insulating housing protecting this electrical conductor, should be prevented. Standards, such as International Electrotechnical Commission Standard IEC 60529, require a protection (for example the so-called IPXXB protection). To meet the requirements of the IPXXB protection for example, a test joint finger of 12 mm diameter and 80 mm length should have an adequate clearance from hazardous parts. Such a requirement limits the diameter of a cavity or passage of a housing opened on an energized electrical conductor.

(4) However, when connectors are used in high current intensity applications, heating should be limited. Therefore, there is a need for power terminals having a large cross section. But large cross-sections may not be compatible with IPXXB protection when power terminals have to be inserted through a cavity or passage of an insulating housing.

(5) This disclosure aims at providing a connector assembly that is compatible with both an IPXXB protection and an electrical power conduction.

SUMMARY

(6) This aim is at least partially achieved by the electrical connector assembly described herein. Indeed, thanks to such an electrical connector assembly, it is possible to use terminals with cross-sections that can correspond to diameters greater than 12 mm, so as to carry high-power currents, while limiting heating, but protection structure(s) create(s) an obstacle for a finger materialized by a cylindrical object of revolution with a diameter greater than, or equal, to 12 mm. This obstacle can be formed by only one protection structure or by a plurality of protection structures. The protection structures can be arranged in many ways. They can be arranged at the same level in the cavity with regard to the insertion direction. Alternatively, they can be arranged at various levels in the cavity with regard to the insertion direction.

(7) According to one or more aspects of the present disclosure, an electrical connector includes a first housing part defining at least one cavity extending longitudinally along an insertion direction; a power terminal with a connection portion configured to be inserted, in the insertion direction, in the at least one cavity, the power terminal having an external and an internal cylindrical surface; and a first electrical conductor at least a portion of which is covered by the first housing part, and configured to be connected to the power terminal through the at least one cavity, when the power terminal is inserted in the at least one cavity, wherein the cavity has an internal surface from which protrudes at least one protection structure, wherein the at least one protection structure is

configured to prevent the insertion, in the insertion direction, in the cavity, of a solid cylinder having a diameter equal to or greater than 12 mm, past the at least one protection structure, wherein at least one slot is defined in the external cylindrical surface, wherein the external cylindrical surface has a diameter greater than 12 mm, wherein each slot extends longitudinally parallel to the insertion direction and extends radially only over a portion of a thickness of the terminal between the external cylindrical surface and the internal cylindrical surface, and wherein the at least one protection structure can slide parallel to the insertion direction in at least one slot.

(8) In some aspects of the electrical connector described in the preceding paragraph, the connection portion defines a hollow portion extending parallel to the insertion direction.

(9) In some aspects of the electrical connector described in any one of the preceding paragraphs, the electrical connector further includes a bolt with a head and a rod, wherein the rod is engaged through the hollow portion and is screwed.

(10) In some aspects of the electrical connector described in any one of the preceding paragraphs, the electrical connector further includes a second housing part covering at least a portion of a second electrical conductor which is electrically and mechanically connected to the first electrical conductor with the power terminal and the bolt.

(11) In some aspects of the electrical connector described in any one of the preceding paragraphs, the first electrical conductor is a busbar.

(12) In some aspects of the electrical connector described in any one of the preceding paragraphs, each protection structure is a rib extending parallel to the insertion direction and wherein each slot is configured to receive a rib.

(13) In some aspects of the electrical connector described in any one of the preceding paragraphs, the at least one slot includes a plurality of slots symmetrically arranged around a central axis which is parallel to the insertion direction.

(14) In some aspects of the electrical connector described in any one of the preceding paragraphs, the at least one cavity and the at least one protection structure are configured to meet International Electrotechnical Commission Standard IEC 60529 with regard to IPXXB protection.

(15) In some aspects of the electrical connector described in any one of the preceding paragraphs, the external cylindrical surface extends over a cylinder having a diameter of 17.1 mm.

(16) In some aspects of the electrical connector described in any one of the preceding paragraphs, the at least one cavity has an internal surface from which three ribs forming the at least one protection structure extend, wherein each rib terminates with a ridge arranged towards a central axis of the at least one cavity, wherein the ridges of the three ribs are arranged in a circle extending perpendicularly to the central axis, and wherein the diameter of the circle is less than or equal to 12 mm.

(17) In some aspects of the electrical connector described in any one of the preceding paragraphs, each rib has an upper end which is at least at a distance of 43.2 mm from the first electrical conductor or any other electrically conducting part in electrical contact with the first electrical conductor and wherein the upper end is visible through a first cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention is now described, by way of example with reference to the accompanying drawings, in which:

(2) FIG. 1 is a schematic exploded view in perspective an example of an embodiment of a connector assembly according to some embodiments;

(3) FIG. 2 shows schematically in perspective, a first housing part of the connector assembly shown in FIG.1 according to some embodiments;

- (4) FIG. 3 shows schematically in perspective, a second housing part of the connector assembly shown in FIG. 1 according to some embodiments;
- (5) FIG. 4 shows schematically in perspective, a power terminal for the connector assembly shown in FIG. 1 according to some embodiments;
- (6) FIG. 5 is a schematic longitudinal cross-section of the connector assembly shown in FIG. 1 according to some embodiments;
- (7) FIG. 6 shows schematically in perspective, with a test finger inserted in a cavity of the first housing part of the connector assembly shown in FIG. 1 according to some embodiments; and
- (8) FIG. 7 is a schematic longitudinal cross-section of a detail of the connector assembly shown in FIG. 1, and a test finger inserted in one of its cavities according to some embodiments.

DETAILED DESCRIPTION

- (9) An example of an electrical connector assembly **1** for an electrical power circuit of a motor vehicle is shown in FIG. 1. The electrical connector assembly **1** comprises, amongst other elements, a first housing part **2**, a second housing part **3**, a power terminal **4** (not shown in FIG. 1), bolts **5**, a first electrical conductor **6** and a second electrical conductor **7**.
- (10) The electrical connector assembly **1** may also comprise for example a cover **29**, an upper interface housing part **8**, an environment part **9** (onto which the electrical connector assembly **1** is attached), a cap **10** (for protecting the first electrical conductor **6** from dust), a thermal sensor assembly **11** (connected to a circuit **12**), and a bottom housing part **13**.
- (11) The first housing part **2** and the second housing part **3** are made of electrically insulating material. For example, they are made of molded plastic.
- (12) As shown in FIG. 2, the first housing part **2** comprises a plurality of first cavities **14**. Each first cavity **14** extends along an insertion direction I. In the illustrated example, the first housing part **2** comprises two cavities **14**. Each first cavity **14** has a general cylindrical shape of revolution around a central axis A. The central axis A is parallel to the insertion direction I. Each first cavity **14** has an internal diameter greater than 17.1 mm. Each first cavity **14** has an internal surface from which extends a plurality of ribs **16** forming protection structures. In the illustrated example, each first cavity **14** comprises three ribs **16** symmetrically arranged, with a 120-degree angle, around the central axis A. Each rib **16** extends longitudinally parallel to the insertion direction I. Each rib **16** extends radially towards the central axis A. Each rib **16** terminates towards the central axis A by a ridge **17**. The ridges **17** of the three ribs **16** are arranged in a circle extending perpendicular to the central axis A. The diameter of this circle is less or equal to 12 mm. The upper end **18** of each rib **16** is at least at a distance D of 43.2 mm from the first electrical conductor **6** or any other electrically conducting part in electrical contact with the first electrical conductor **6** and that is visible through a first cavity **14** (see FIG. 5).
- (13) As shown in FIG. 3, the second housing part **3** comprises a plurality of second cavities **15**. Each second cavity **15** extends along the insertion direction I. In the illustrated example, the second housing part **3** comprises two second cavities **15**. Each second cavity **15** has a general cylindrical shape of revolution around the central axis A. Each second cavity **15** has an internal diameter greater than 17.1 mm. Each second cavity **15** has an internal surface from which extends a plurality of pins **19**. The plurality of pins **19** form coding structures. In the illustrated example, each second cavity **15** comprises three pins **19** symmetrically arranged, with a 120-degree angle, around the central axis A. Each pin **19** extends radially towards the central axis A. Each pin **19** terminates towards the central axis A by a free end **20**. The free ends **20** are arranged in a circle extending perpendicular to the central axis. The diameter of this circle is greater or equal to 12 mm.
- (14) The power terminal **4** is made of electrically conducting material. As shown in FIG. 4, the power terminal **4** has a general cylindrical shape forming a connection portion and extending longitudinally parallel to the insertion direction I. The power terminal **4** has an external cylindrical surface **21** and an internal cylindrical surface **22**. The external **21** and internal **22** surfaces are symmetrical around the central axis A. The internal cylindrical surface **22** defines a hollow portion

25. The external cylindrical surface **21** extends over a cylinder of 17.1 mm diameter. The external cylindrical surface **21** defines slots **23**. In the illustrated example, the external cylindrical surface **21** defines three slots **23** symmetrically arranged, with a 120-degree angle, around the central axis A. Each slot **23** extends longitudinally parallel to the insertion direction I. Each slot **23** extends radially towards the central axis A. Each slot **23** is large enough (circumferentially) and deep enough (radially) for accommodating a first rib **16** or pin **19**. The power terminal **4** also has a collar **24** extending radially from the external cylindrical surface **21**. For connecting the first **6** and second **7** electrical conductors, the power terminal **4** is first inserted into the second cavity **15** of the second housing part **3**. During this insertion, the pins **19** serve a coding means for aligning the slots **23** with the ribs **16**.

(15) As illustrated in FIG. 5, the bolt **5** has a rod **26** and a head **27**. The rod **26** and the head **27** are made of an electrically conducting material. The head **27** is covered with a protective cap **28** made of an electrically insulating material. The rod **26** is inserted in the hollow portion **25**.

(16) In the illustrated example, the first electrical conductor **6** and a second electrical conductor **7** are busbars. The first electrical conductor **6** and second electrical conductor **7** may be included in an electrical power circuit of a vehicle. The first electrical conductor **6** and second electrical conductor **7** may be made for example of aluminum or copper (aluminum or copper alloy) and covered by an insulating material. A portion of the first electrical conductor **6** is covered by the first housing part **2**. A portion of the second electrical conductor **7** is covered by the second housing part **3**. A nut **30** is screwed onto the rod **26**. A first electrically conductive ring **31** is engaged and blocked into a hole made in the first electrical conductor **6**. The head **27** engages a second electrically conductive ring **32** that itself is engaged and blocked into a hole made in the second electrical conductor **7**. When the bolt **5** is tightened, the power terminal **4** is in electrical contact with at least the first electrically conductive ring **31** and the second electrically conductive ring **32**, that are themselves electrical contact respectively in with the first electrical conductor **6** and the second electrical conductor **7**. The first **6** and second **7** electrical conductors are therefore electrically connected through a round power terminal **4** having a diameter which is greater than 12 mm.

(17) A test finger **33** is shown in FIG. 6. The test finger **33** has a general cylindrical shape forming a solid cylinder (terminated by tapered tip **34**) having an external diameter of 12 mm. Therefore, the test finger **33** is blocked by the upper ends **18** of the ribs **16** and is therefore prevented from touching the first electrical conductor **6**.

(18) As illustrated in FIG. 7, the tip **34** of the test finger **33** is at a distance L of at least 10.5 mm from the first electrically conductive ring **31**.

(19) Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the following claims, along with the full scope of equivalents to which such claims are entitled.

(20) As used herein, 'one or more' includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

(21) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

(22) The terminology used in the description of the various described embodiments herein is for the purpose of describing embodiments only and is not intended to be limiting. As used in the

description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes”, “including”, “includes”, and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(23) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

(24) Additionally, while terms of ordinance or orientation may be used herein these elements should not be limited by these terms. All terms of ordinance or orientation, unless stated otherwise, are used for purposes distinguishing one element from another, and do not denote any order of arrangement, order of operations, direction or orientation unless stated otherwise.

Claims

1. An electrical connector assembly for an electrical circuit of a motor vehicle, comprising: a first housing part defining at least one cavity extending longitudinally along an insertion direction; a power terminal with a connection portion configured to be inserted, in the insertion direction, in the at least one cavity, the power terminal having an external and an internal cylindrical surface; and a first electrical conductor at least a portion of which is covered by the first housing part, and configured to be connected to the power terminal through the at least one cavity, when the power terminal is inserted in the at least one cavity, wherein the cavity has an internal surface from which protrudes at least one protection structure, wherein the at least one protection structure is configured to prevent the insertion, in the insertion direction, in the cavity, of a solid cylinder having a diameter equal to or greater than 12 mm, past the at least one protection structure, wherein at least one slot is defined in the external cylindrical surface, wherein the external cylindrical surface has a diameter greater than 12 mm, wherein each slot extends longitudinally parallel to the insertion direction and extends radially only over a portion of a thickness of the terminal between the external cylindrical surface and the internal cylindrical surface, and wherein the at least one protection structure can slide parallel to the insertion direction in at least one slot.
2. The electrical connector assembly according to claim 1, wherein the connection portion defines a hollow portion extending parallel to the insertion direction.
3. The electrical connector assembly according to claim 2, further comprising a bolt with a head and a rod, wherein the rod is engaged through the hollow portion and is screwed.
4. The electrical connector assembly according to claim 3, further comprising a second housing part covering at least a portion of a second electrical conductor which is electrically and mechanically connected to the first electrical conductor with the power terminal and the bolt.
5. The electrical connector assembly according to claim 1, wherein the first electrical conductor is a busbar.
6. The electrical connector assembly according to claim 1, wherein each protection structure is a rib extending parallel to the insertion direction and wherein each slot is configured to receive a rib.
7. The electrical connector assembly according to claim 1, wherein the at least one slot comprises a plurality of slots symmetrically arranged around a central axis which is parallel to the insertion direction.

8. The electrical connector assembly according to claim 1, wherein the at least one cavity and the at least one protection structure are configured to meet International Electrotechnical Commission Standard IEC 60529 with regard to IPXXB protection.

9. The electrical connector assembly according to claim 1, wherein the external cylindrical surface extends over a cylinder having a diameter of 17.1 mm.

10. The electrical connector assembly according to claim 1, wherein the at least one cavity has an internal surface from which three ribs forming the at least one protection structure extend, wherein each rib terminates with a ridge arranged towards a central axis of the at least one cavity, wherein the ridges of the three ribs are arranged in a circle extending perpendicularly to the central axis, and wherein the diameter of the circle is less than or equal to 12 mm.

11. The electrical connector assembly according to claim 10, wherein each rib has an upper end which is at least at a distance of 43.2 mm from the first electrical conductor or any other electrically conducting part in electrical contact with the first electrical conductor and wherein the upper end is visible through a first cavity.
